Self-Awareness and Selfhood in Human and AI Cognition: Toward a Future Architecture for the Formation of High-Context Intelligence

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Abstract: This paper presents a theoretical framework for reconsidering the nature of human intelligence and selfhood in order to clarify the structural limitations of contemporary artificial intelligence and the conditions under which future AI systems may transcend them. Human selfhood is not generated by a conscious internal observer, but emerges from a hierarchical predictive structure integrating unconscious processes, retrospective narrative construction, and self-maintenance of the organism.

From this perspective, current AI foundation models possess sophisticated predictive capabilities and can generate contextually coherent representations; however, they lack the embodied, self-maintaining, and hierarchically integrated architecture underlying human self-models. Therefore, the limitations of AI should not be understood merely as problems of computational performance, but as questions concerning the ontological conditions that enable meaning, cognition, and social understanding.

This paper further distinguishes between functional approximation and ontological participation, arguing that the external reproduction of human-like cognitive and social behaviors alone does not imply access to the constitutive background from which meaning emerges. It suggests that overcoming these limitations would require a fundamentally different form of artificial existence based on self-maintaining, biologically grounded intelligence. Therefore, while future AI systems may acquire increasingly sophisticated forms of self-awareness, such advances alone would not entail the emergence of selfhood.

Keywords: Artificial Intelligence, Self-Model, Predictive Processing, High-Context Understanding, Representational Computation, Functional Approximation, Ontological Participation, Metaphysical Layer

Section-I. A Structural Theory of High-Context Meaning

Recent advances in deep neural-network-based AI have dramatically expanded the capacity of machines to extract statistical regularities and to generate contextually plausible responses. Despite these achievements, such systems remain confined to low-context forms of coherence, fundamentally differing from the high-context meaning structures upon which human societies have historically depended. Existing discussions of "high-context," most notably E. T. Hall's theory of intercultural communication, define the term primarily in relation to implicit communication and shared background knowledge. While valuable within its original domain, Hall's interpretation does not capture the deeper causal, institutional, and normative structures through which human beings construct and negotiate meaning.

The term "high-context" as employed in this paper therefore diverges in conceptual scope from Hall's formulation. Whereas Hall's framework concerns communicative implicitness, the present usage designates a structural domain of meaning encompassing value hierarchies, institutional background, historical situatedness, and culturally embedded systems of responsibility. High-context is not reducible solely to tacit knowledge, reading between the lines, or implicit mutual understanding. Rather, it is defined as a semantic space in which what is not explicitly stated acquires meaning, a domain that presupposes counterfactual evaluation of unselected actions and incorporates purpose hierarchies, reciprocal expectations of responsibility, and institutionalized value structures as essential causal components. In this sense, "high-context" refers not merely to what is said, but to the broader semantic structure through which what remains unsaid becomes meaningful.

High-context recognition certainly exists in everyday life. In linguistic expressions, "I remembered to lock the door" implies that the act of locking the door was performed as the result of selecting it from among multiple possible courses of action, such as leaving without locking the door or entrusting a housemate with locking it. In contrast, "I remembered locking the door" merely expresses the recollection of the past event of having locked the door and does not indicate anything about the decision-making process through which that action was carried out. For this reason, compared with "I remembered locking the door", "I remembered to lock the door" represents a higher level of contextual representation because it incorporates the context of action selection.

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The same structural principles can also be identified in the operational logs generated by vehicle safety systems. Vehicle safety systems can record the occurrence of an action (the event), but the causal structure that governed the selection of that action is not directly observable from those records. In other words, operational logs document what happened, yet they provide no direct access to the underlying causal rationale, the latent evaluative states, counterfactual exclusions, and purpose-driven pathways through which the system arrived at that particular action. Hence, the mere availability of observable logs does not entail observability of the system's internal causal structure.

Observable logs $\neq$ Causal-structure observability.

This linguistic distinction reveals a broader structural principle: meaning is not determined solely by the representation of an event, but also by the implicit relations among intention, selection, and situational constraints that constitute the background of that event. Thus, contextual information is embedded not only in the observable action itself, but also in the structure of alternatives and conditions through which that action becomes meaningful.

This structural principle is directly relevant to the interpretation of AI systems and vehicle safety logs. A record indicating that a particular action occurred captures only the observable outcome of a decision process. However, a higher level of contextual representation requires information regarding the "conditions" under which the action was selected, the alternative actions available at that moment, and the rationale underlying the prioritization of one action over others.

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This distinction becomes particularly significant in the context of Level-4 ADS (Automated Driving System) when interpreting MRMs (Minimum Risk Manoeuvres).
UN Regulation No. 157 (WP.29) explicitly refers to an MRM lane change. In particular, paragraph 5.5.1 requires that an MRM bring the vehicle to a standstill in a target stop area considered to achieve the greatest minimisation of risk under the given circumstances and, where necessary, perform an MRM lane change if the vehicle is capable of doing so.

Here, in the UNECE 2026 ADS Regulation, "ADS" stands for Automated Driving System, not Autonomous Driving System.

The provision concerning the target stop area in No. 157 is as follows: This shall be in a target stop area considered to be the greatest minimisation of risk achievable under the given circumstances. It does not merely require the vehicle to stop before an obstacle; rather, it requires the vehicle to stop in a target stop area considered to achieve the greatest minimisation of risk under the given circumstances. However, the recorded manoeuvre alone does not reveal the rationale for selecting that manoeuvre, or target stop area, was selected over other feasible alternatives. Consequently, event recording alone may be insufficient for causal explanation and responsibility attribution.

Ref: **UN Regulation No.157 (WP.29), paragraph 5.5**
The minimum risk manoeuvre shall bring the vehicle to a standstill unless the system is deactivated by the driver during the manoeuvre. This shall be in a target stop area considered to be the greatest minimisation of risk achievable under the given circumstances (e.g. traffic situation, environmental conditions, system failures), performed according to paragraph 5.2.6, if a lane change is required to reach the target stop area and the ALKS (Automated Lane Keeping Systems) is capable of performing an MRM lane

change. Otherwise, within its current lane, or in the case the lane markings are not visible, following an appropriate trajectory taking into account surrounding traffic and road infrastructure.

1. The MRM (Minimum Risk Manoeuvre) shall bring the vehicle to standstill.

   Note-1: The term "minimum" in MRM does not denote a minimum level of manoeuvring effort. Instead, it refers to achieving the greatest practicable minimisation of risk under the prevailing circumstances.

   On the other hand, this interpretation remains valid and is unchanged by the 2026 ADS regulation. R157 continues to define MRM only for ALKS, not for ADS.

2. The vehicle shall come to a standstill in a target stop area.

   Note-2: The target stop area concept does not imply an emergency stop followed by subsequent vehicle movement to reach the target stop area.

   R157 does not explicitly prohibit an emergency stop followed by subsequent movement. It only requires that the final standstill occur in a target stop area representing the greatest minimisation of risk.

   Nonetheless, this remains an ALKS-specific requirement; ADS vehicles follow the Fallback (Level-3 ADS)/MRC (Mitigated Risk Condition) procedures defined in the 2026 ADS regulation, not R157.

3. The target stop area shall be considered to represent the greatest minimisation of risk achievable under the given circumstances (e.g. traffic situation, environmental conditions, system failures).

   Note-3: The target stop area shall be the area in which the achievable risk is minimized under the prevailing conditions.

   This definition remains valid for ALKS. However, under the ADS Regulation adopted by WP.29 in June 2026, ADS perform risk-mitigation behaviour within the separate framework of the MRC.

   Thus, the risk-mitigation behaviour of ADS does not follow the R157 concept of "a target stop area" but instead complies with the MRC/MRM-equivalent concepts defined in the ADS Regulation.

4. If a lane change is required to reach the target stop area and the ALKS is capable of performing an MRM lane change, the MRM lane change shall be performed.

   Note-4: UN Regulation No. 157 prescribes MRM requirements only for ALKS.

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ADS related MRM/MRC concepts are addressed in the separate ADS regulation adopted by WP.29 in June 2026. Accordingly, the WP.29 (Post-June-2026) ADS Regulation defines safety-assuring behaviour functionally equivalent to the MRM/MRC concepts.

1. The ADS shall be capable of achieving and maintaining a MRC in the event of a system failure, operational design domain exit, or other triggering conditions.
2. The ADS shall perform risk-mitigation behaviour that brings the vehicle to, and maintains it in, a state that minimizes risk to occupants and other road users.

Specifically, the difference between the "target stop area" defined in UN Regulation No. 157 and the "Mitigated Risk Condition (MRC)" specified in the ADS Regulation is as follows. R157 (ALKS) requires the vehicle to reach and stop in a "target stop area," defined as the location that represents the greatest minimisation of risk under the prevailing circumstances. This requirement specifies a standstill as the final state of the Minimum Risk Manoeuvre.

In contrast, the ADS Regulation adopted by WP.29 in June 2026 requires an ADS to achieve and maintain a MRC. Under the ADS Regulation, risk-mitigation behaviour does not necessarily involve bringing the vehicle to a standstill. The ADS may continue moving at a safe speed or follow a safe trajectory when this results in a lower overall risk. Accordingly, ADS risk-mitigation behaviour follows the MRC framework defined in the ADS Regulation and does not rely on the R157 concept of a "target stop area."

Unlike R157, which defines a specific physical location where the vehicle must come to rest, the ADS Regulation defines MRC as "a vehicle state" rather than a geographical destination. MRC is satisfied when the ADS maintains a condition in which risk is minimized, regardless of whether the vehicle is stationary or moving.
Therefore, the ADS Regulation separates the concept of risk mitigation from the requirement to reach a particular stopping location. The regulatory focus is on achieving "a safe operational state", not on arriving at a predetermined stop area.
Furthermore, the ADS Regulation treats risk-mitigation behaviour as "a continuous process" rather than "a discrete manoeuvre". While R157 prescribes a single manoeuvre culminating in a standstill, the ADS Regulation requires the ADS to maintain the vehicle in an MRC as conditions evolve, which may involve controlled motion, lane positioning, or other actions that minimise "risk" without requiring a final stop.

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Consequently, R157 requires the ALKS to select a context-dependent "target stop area", and, analogously, the ADS regulations also require the ADS to achieve and maintain a "Mitigated Risk Condition (MRC)". However, observable event logs alone cannot reconstruct the causal reasoning underlying the system's achievement and maintenance of the MRC, because the concept of the MRM/MRC presupposes "high-context" information.

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WP.29 requires that the safety of an ADS be demonstrated through a Safety Case during the approval process. However, it does not merely require the submission of a Safety Case. Rather, it explicitly specifies the structural requirements necessary to support that safety argument, including the Safety Management System (SMS), the Data Storage System for Automated Driving (DSSAD), and In-Service Monitoring and Reporting. In other words, WP.29 itself adopts a regulatory framework based on the principle that safety claims must be supported by explicit structural requirements.

Here, the Safety Case represents the objective of demonstrating safety claims, and its structural elements constitute the means necessary to support that objective. That is, the institutional design adopted by WP.29 is based on the principle that any explanation presented in a Safety Case must be supported by substantive structural requirements.

On the other hand, WP.29 defines the MRC as a central safety concept for ADS and requires that an ADS be capable of achieving and maintaining an MRC. However, with respect to demonstrating, after an accident, that the ADS had achieved and maintained an MRC, WP.29 does not explicitly specify structural requirements comparable to those required at the approval stage, such as the SMS or DSSAD.

The claim that the regulatory design adopted by WP.29, as a matter of international institutional design, is not positioned to explicitly specify the substantive structural requirements (the means) necessary to support accountability for post-accident explanations concerning the achievement and maintenance of the MRC (the objective) does not appear to be consistent with the regulatory design adopted by WP.29 during the approval stage. This is because that regulatory design explicitly specifies not only the requirement to provide a Safety Case (the objective) but also the structural requirements (the means) necessary to support that objective as part of the approval stage.
In other words, although WP.29 adopts a regulatory framework in which a Safety Case must be supported by explicit structural requirements during the approval stage, it does not specify comparable structural requirements for demonstrating, after an accident, that an ADS had achieved and maintained an MRC.

The logical structure of the argument can be summarized in the following four steps:

1. During the approval stage, WP.29 explicitly specifies the substantive structural requirements necessary to support the Safety Case, including the SMS, the DSSAD, and In-Service Monitoring and Reporting.
2. In other words, WP.29 adopts a regulatory design based on the principle that safety-related explanations must be supported by explicit substantive structural requirements.

3. Furthermore, WP.29 defines the MRC (the objective) as a central safety concept for ADS and requires that an ADS be capable of achieving and maintaining an MRC.
4. Nevertheless, if it is argued that the substantive structural requirements (the means) necessary to support accountability for post-accident explanations concerning the achievement and maintenance of the MRC (the objective) fall outside the scope or regulatory purpose of WP.29, such a position does not appear to be consistent with the regulatory design adopted by WP.29 itself as part of its type-approval system.

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At this point, the central points of the argument can be organized as follows: If, in the event of an accident, it is possible to demonstrate the achievement and maintenance of the MRC by using the structural requirements that support the Safety Case, such as the Safety Management System (SMS), the Data Storage System for Automated Driving (DSSAD), and In-Service Monitoring and Reporting, then no problem arises.

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Why observable event logs alone cannot reconstruct the causal reasoning underlying an ADS's achievement and maintenance of the MRC.

This is because the very concept of an MRC presupposes "high-context" information that may not be reducible to recorded system events.

In order to deepen the conceptual understanding of this issue, it is necessary to recognize that contemporary DNN-based AI systems lack the architectural mechanisms required to access such high-context domains. They cannot explicitly represent purpose hierarchies, cannot generate counterfactual assessments of actions that were not selected, and provide no means of specifying the admissible alternatives from which a particular action emerged. Moreover, they possess no architectural mechanisms capable of recording or explicating how value judgments are prioritized (see Section-V Note-2,-3).

As a result, current AI systems can produce statistically coherent and contextually plausible responses, but they cannot construct the culturally and institutionally embedded meaning structures that constitute high-context understanding.

Understanding the reasons for this limitation requires an integrated analysis across multiple interrelated levels, encompassing not only brain function but also immune function and psychoanalytic conceptions of the unconscious.

This analysis suggests that future AI-enabled societies will require an institutional surrogate capable of compensating for AI's structural inability to engage with "high-context" meaning.

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Following the June 2026 regulatory revision, WP.29 adopted the institutional principle that any explanation must be supported by the substantive structural requirements necessary to sustain it. Explanations concerning the post-accident achievement and maintenance of the MRC likewise necessarily encompass “high-context” information and therefore require comparable supporting structures. However, such structures do not exist within the internal architecture of contemporary AI systems. Consequently, they must instead be provided by the institutional framework itself. This is the essential role of the institutional surrogate.

Section-II. Biological Immunity and Consciousness as an Individual

To define oneself as an existent presupposes two foundational grounds: the intentional activity of consciousness, “self-awareness”, constituted within the brain, and the immune function that safeguards the organism’s embodied unity.

Note-1: “self-awareness” refers to the capacity to recognize oneself as an object of one’s own attention. "selfhood" refers to the constitution of the self as an existing entity.

The immune system performs the fundamental functions of distinguishing self from non-self, recognizing pathogenic or function-disruptive agents, initiating effector responses against such threats, and establishing immunological memory that enables more rapid and effective secondary responses. Collectively, these processes constitute biological mechanisms directed toward the maintenance of organismal integrity and, from a phylogenetic perspective, may be regarded as representing one of the earliest manifestations of biological autonomy.

The intestinal tract, particularly the large intestine, serves as the principal site of immune regulation and is thought to begin developing during embryogenesis before the central nervous system. Embryonic development proceeds in the following sequence: (1) formation of the primitive gut tube, (2) cardiac morphogenesis, and (3) development of the neural tube, from which the brain and spinal cord subsequently arise.

This developmental sequence indicates that the physiological systems responsible for nutrient processing and immune defense are established before the maturation of the central nervous system. Interpreted philosophically, life first establishes the capacities for digestion and immune defense, and only afterward constructs the brain.

Note-2: Approximately 70–80% of the body’s immune cells are thought to reside in the intestinal tract, with a substantial proportion of gut-associated immunological activity occurring in the large intestine. The colonic mucosa contains high densities of IgA-producing plasma cells, regulatory T (Treg) cells, and dendritic cells. Furthermore, short-chain fatty acids (SCFAs), particularly butyrate, produced through microbial fermentation in the colon, play central roles in immunomodulation. Collectively, these observations support the view that the large intestine constitutes the principal site of gut-associated immune regulation.

The immune system does not merely execute predetermined biochemical reactions. It continuously monitors internal and external conditions, discriminates among innumerable molecular patterns, predicts biological threats on the basis of prior immunological memory, and adaptively selects appropriate responses.
In this functional sense, immunity embodies information processing, evaluation, memory, and decision-making independently of conscious cognition. This constitutes a functional capacity that exceeds the information-processing procedures implemented by the most advanced AI systems today.
On this basis, it is concluded that the essence of intelligence does not reside solely in the conscious processes of the brain, but also in the body itself, which possesses biological autonomy as an individual organism.
Conscious "self-awareness" therefore emerged on the foundation of a pre-existing biological system in which "spontaneous self-organization", "self-preservation", and environmental evaluation were already established. It did not precede that system. Hence, intelligence cannot reasonably be regarded as an exclusive property of the brain.

In psychoanalytic terms, prior to the emergence of linguistically expressible consciousness, the brain contains "countless latent cognitive patterns" that have been formed through continuous interactions with other people and the surrounding environment throughout past experience. Most of these patterns remain outside conscious "self-awareness" and constitute the domain of the unconscious. When confronted with a particular situation, the brain selects, integrates, and reorganizes those patterns of thought that are most adaptive for maintaining the mode of existence of the self under the circumstances.(see Section-III)

The resulting configuration is then manifested as linguistically expressible conscious "self-awareness". Hence, bodily "self-preservation", unconscious "cognitive organization", and conscious "self-awareness" constitute three interdependent layers of human intelligence rather than independent processes. From this perspective, it is reasonable to regard the unconscious as a vast repertoire of thought patterns accumulated for the preservation of the self, whereas consciousness represents the temporary configuration that emerges through the selection and integration of those patterns in accordance with the demands of a given situation.

To further clarify the relationship between the intestine and the brain, it is worth emphasizing the following. Given the immunological architecture concentrated in the

intestine, a perspective emerges in which bodily autonomy is grounded not in the brain but in the intestinal system itself.
Peyer's patches, GALT, IgA-producing cells, dendritic cells, macrophages, T cells, and the vast intestinal microbiota collectively constitute a boundary interface with the external environment, continuously processing information and maintaining physiological homeostasis. Moreover, the adult human brain contains about 86 billion neurons, and the intestine likewise possesses about 500 million neurons. This consideration leads to the following bold hypothesis: it is conceivable that the intestine governs the primary form of radical intelligence, whereas the brain constitutes a secondary, derivative form of intelligence.

This organization suggests a functional continuity in which the intestine constitutes the biological foundation of immunity, immunity provides the foundation for "autonomy", and "autonomy" provides the foundation for intelligence. Although the brain later evolved into a highly integrated control apparatus, many foundational processes underlying biological evaluation, reaction, and adaptation have long been carried out spontaneously by the intestinal immune network, without external control or compulsion. From this perspective, intelligence may be viewed as having evolved from biological systems of "self-preservation" centered on intestinal immunity.

*Autonomy (noun): the ability to govern oneself or control one's own affairs.
(adjective): Autonomous

*Spontaneity (noun): the quality or state of occurring or acting naturally, without being forced or planned. (adjective): Spontaneous

Ref-1: **The Anthropic Principle and the Mythos in Relation to AI and Security**
The anthropic principle, which in cosmology refers to the "observation" that the physical laws and constants of the universe appear finely tuned to permit "human existence", acquires a broader philosophical implication when considered in relation to meaning.
It develops into a perspective that necessarily incorporates the conditions of "human existence": the very fact that "observers" exist becomes the premise upon which structures of meaning are built.

Scientifically, it is generally accepted that the universe exists independently of human beings. However, to pose questions such as Was the universe created for the sake of humanity?, Why does the universe exist?, or Does the universe possess purpose or meaning? is itself to inquire into how the universe should be understood and what significance should be attributed to it. Such questions can be raised only under the precondition that human observers exist. Therefore, the question of the existential significance of the universe is distinct from the question of the universe's objective existence; rather, it constitutes a philosophical inquiry that arises only under the precondition that human observers exist. In the absence of human beings, although the universe itself would still exist, there would be no subject capable of asking what significance, if any, the universe possesses.

If a degree of philosophical speculation is permitted, the implications of the present discussion can be extended further. It can be argued that ontological inquiries into the universe appear, at a superficial level, to concern the meaning of the universe itself. However, the most fundamental premise underlying the possibility of such inquiries is the existence of the questioning subject, the self who poses the question. In this respect, the present discussion reveals a philosophical structure analogous to Descartes' cogito, which established the existence of the thinking subject as the starting point of philosophy.

"Self-awareness" emerges only through the existence of others. However, others do not exist for the sake of the self. By the same reasoning, the universe does not exist for the sake of humanity, nor does humanity exist for the sake of the universe.
It seems that the anthropic principle contains a distinctly philosophical "high-context" dimension: namely, that "self-awareness" develops only in relation to others, just as the significance of the universe can be articulated only under the precondition that human observers exist. Stated more strongly, just as the significance of the universe presupposes the existence of human observers, "self-awareness" is possible only in relation to others.

In other words, it may be argued that participation in the universe as an existent, namely ontological participation, constitutes the necessary condition for the emergence of meaning, selfhood, and high-context understanding.

Ref-2: **Conceptual Positioning of Mythos**
The notion of mythos in this context does not denote mere "mythology". It signifies a framework through which "human existence" is represented and stabilized by institutions, culture, and narrative. The concept of the "Anthropic Mythos" proposed in this paper arose from discussions on AI safety and AI security, where the need to reconsider the conditions of human existence from institutional and narrative perspectives has become increasingly apparent.

Ref-3: **Structural Analogy Between Biological Immunity and Technological Security**
Specifically, the analogy between biological immunity and technological security can be understood as indicative of a deeper structural essence. The intestinal immune system monitors boundaries, prevents intrusion, and retains memory of past encounters. In parallel, cybersecurity, including AI system security, detects external attacks, blocks them,

and preserves a history of such attacks in the form of logs.
In this sense, immunity can be regarded as a biological form of security, whereas AI safety, represents a technological form of security.

Ref-4: **Structural Homology and the Anthropic Mythos**

This analogy suggests a structural "homology": human existence is conditioned by biological immunity, whereas social and technological existence is conditioned by security architectures. Recognition of this homology carries direct implications for institutional design in societies mediated by AI. It is precisely the linkage between biological conditions of existence and technological conditions of security that elucidates the concept of the "anthropic mythos".

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In this paper, it is hypothesized that Ontological Participation generates individualized meaning through the process by which an agent interacts with the environment as a situated embodied entity within the world. Furthermore, this paper assumes that, simultaneously with the generation of individualized meaning, within this participatory process, individualized qualia arising through the agent's own biological sensory organs are integrated with the generated individualized meaning, thereby forming a self-referential representational structure constituting selfhood and ultimately leading to the emergence of a Self-model. Biological Autonomy(Immune self/non-self distinction) → Ontological Participation → Generation of Individualized Meaning "**+**" Individualized Qualia → Integration into a Self-referential Structure → Emergence of a Self-model

Section-III. The Conceptual Trajectory of the Self-Model

In research on brain function, Gazzaniga's seminal contribution was the demonstration that the left hemisphere functions not as a device for reporting facts but as a mechanism for generating narratives. In split-brain patients, the right hemisphere can initiate behavior in response to stimuli that the left hemisphere never receives. Although the left hemisphere lacks access to the actual cause of the action, it nonetheless produces a coherent and seemingly rational explanation after the fact. This finding suggests that the reasons we ordinarily experience as "our own decisions" are, at least in part, post hoc constructions generated only after the behavior has already occurred.

This interpretation was reinforced by Libet's experiment, which showed that neural activity associated with voluntary movement begins several hundred milliseconds before participants report the conscious intention to act. The decision to initiate movement appears to arise at an unconscious level first, with consciousness registering the decision only afterward and supplying a retrospective rationale. Taken together, Gazzaniga's interpreter hypothesis and Libet's findings challenge the intuitive causal model in which conscious intention precedes and produces action.

Ref-1: **Research Areas of Michael S. Gazzaniga**
Active mainly from the 1960s onward in the United States, Gazzaniga was a pioneering cognitive neuroscientist. His split-brain research revealed that the left hemisphere functions as a "narrative generator," not merely as a fact reporter, profoundly shaping theories of consciousness and self. His research focused on split-brain studies, cognitive neuroscience, and consciousness research.

Ref-2: **Research Areas of Benjamin Libet**
Working primarily in the 1980s, Libet was a neurophysiologist whose experiments on readiness potentials showed that unconscious brain activity precedes conscious intention in voluntary movement. His findings sparked major debates on free will and the temporal structure of consciousness. His research encompassed neurophysiology, consciousness timing, and free will research.

More recent research, however, indicates that the brain does not contain two fully independent personalities, as early interpretations of split-brain data sometimes implied. Although the left hemisphere's tendency to generate post hoc explanations is robust, this mechanism alone cannot account for the full phenomenon of selfhood.
The self arises from a broader and more integrated set of neural processes, not merely from a narrative module.

Against this background, contemporary cognitive science has advanced a powerful theoretical framework known as Predictive Processing. In this view, the brain does not passively "perceive" the external world; rather, it actively "predicts" it. The brain continuously generates expectations about future sensory input and updates these expectations by minimizing the discrepancy between prediction and actual input.

Ref-3: **Structural Analogy Between Human Anticipatory Drive and AI Scaling Laws**
Humans appear to possess an intrinsic "drive" to anticipate the future, and in order to satisfy this fundamental motivational drive, the brain exploits as much past information as possible to generate predictions about an otherwise unknown future.
Similarly, the principle underlying "AI scaling laws" reflects an analogous underlying principle. The belief that larger volumes of past data enable more accurate prediction of an indeterminate future appears to be grounded in a deep motivational disposition, one that is structurally parallel to the human anticipatory "drive" and ultimately rooted in fundamental epistemic and predictive needs (see Afterword). Applied to the notion of self, this framework implies that the self consists of a large-scale predictive model integrating

estimates such as "what kind of person I am," "how I tend to act," and "what I am currently feeling." In other words, the self is an internal model constructed by the brain to forecast its own states and behaviors.

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Predictive processing theory explains the mechanism by which the brain anticipates future input and minimizes the discrepancy between prediction and actual sensory data.
As a theoretical extension, it is natural to infer that humans possess an "innate capacity" to "understand what is not explicitly stated," that is, to recognize and process "high-context" information.

Note-1: An innate capacity refers to a capacity that an organism possesses by virtue of its biological constitution prior to acquisition through learning or experience.

*First, the essence of predictive processing lies in the supplementation of missing information and the resolution of ambiguity through inferential estimation.
*Second, human high-context understanding is closely associated with the ability to infer unspoken social and cultural cues and to adjust behavior and interpretation accordingly.

Thus, when predictive processing is applied to the self-model, its function can be understood as extending beyond the mere prediction of sensory input to encompass the estimation of social context and implicit meaning, thereby contributing to high-context cognition. This line of reasoning suggests that the framework of predictive processing provides a potential foundation for explaining the human capacity for high-context understanding.

At this point, it is instructive to consider that immune function evolved to preserve the organism as an individual entity in the face of disease and infection.
From this perspective, predictive models in cognition can likewise be understood as necessary for preserving the organism under conditions of fundamentally unpredictable future sensory input. The immune system does not "predict" future threats in a computational sense; instead, it maintains a structural capacity to distinguish Self from non-Self and to respond to unknown threats without compromising "organismal integrity". By contrast, the brain's predictive model can be understood as a higher-level analogue of this principle. Just as the immune system provides molecular-level preparedness for maintaining the organism, the brain generates behavioral, affective, and self-referential anticipations that maintain organismal viability under future uncertainty.

In the most recent theories of selfhood, this predictive model is conceived as a dynamic and integrative virtual structure that enables the brain to regulate a complex body within a complex social environment over extended timescales. It is not an illusion, nor is it a “homunculus” observing the world from within the head. It is a functional construct required for the brain to regulate itself.

Note-2: A “homunculus” refers to a hypothetical internal observer or controller residing within the brain that is assumed to perceive the world and direct cognition or behavior. Such explanations simply relocate the explanatory problem to another internal agent, thereby potentially leading to an infinite regress.

From this perspective, Gazzaniga’s interpreter hypothesis, Libet’s evidence for unconscious initiation, the predictive processing framework, and contemporary self-model theory can be regarded as forming a continuous conceptual trajectory. It is thus possible to view these contributions as collectively delineating a progressive line of inquiry into the mechanisms by which unconscious processes, retrospective narrative construction, and predictive integration converge in the constitution of selfhood.

In other words, unconscious processes initiate action; consciousness retrospectively narrates it; and the brain uses these narratives, along with all other available information (including information obtained from sensory organs), to maintain an ongoing predictive model of itself. This predictive structure constitutes the experiential sense of “ I ” and appears to provide the most plausible account of the nature of selfhood.

Ref-4: **Neurological Basis of Predictive Control in Humanoid Robots**

The paper AI Accelerator (ver. 4.5), prepared in September 2025 prior to the present study, discusses "predictive" control in humanoid robots as follows:

The human cerebrum functions as the command center for voluntary movement, setting goals and planning movement trajectories within an abstract space, much like an AI. In contrast, the spinal cord executes selected motor responses under physical and temporal constraints by following instructions issued from the cerebrum.

Distinct from this information-transmission pathway, the cerebellum receives proprioceptive input from the spinal cord via the spinocerebellar tract (SCT).

This input conveys a subset of proprioceptive information concerning the state of muscles and joints, including cutaneous sensation, as well as biological state variables that do not rise to conscious “self-awareness”, such as sudden changes in load, pressure, deformation, temperature, acceleration, vibration, torque fluctuation, and acoustic stimuli.

Using this unconscious information, the cerebellum performs “predictive” and “adaptive” control resembling Model Predictive Control (MPC), continuously forecasting future states while optimizing its neuromotor execution schema (or muscle control signals) in real time.

Through the cerebellum and the spinocerebellar tract (SCT), humans unconsciously and reflexively adjust their own actions.

Through embodied interaction with the environment, humans have evolved not only explicit knowledge but also adaptive heuristics, rules of thumb, or embodied wisdom, that maintain biological integrity by aligning strategic behavior with intrinsic physical constraints.

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When this conceptual trajectory is extended to foundation models, important structural similarities emerge alongside fundamental differences. Transformer architectures internalize statistical regularities in the world through next-token prediction over sequential inputs, thereby generating outputs that are coherent with the surrounding context (see Section-VI and Section-VII). In this respect, they are best regarded as a partial implementation of the predictive-processing architecture. Moreover, much like Gazzaniga's interpreter, their outputs often take the form of coherent post hoc narratives despite the absence of explicit representations of the underlying causal processes.

However, the conditions required for the emergence of a human self-model appear to extend far beyond predictive accuracy alone. They depend on embodiment, homeostatic self-maintenance, responsibility for action selection, and the long-term temporal integration characteristic of a living organism operating within a closed regulatory loop. Although contemporary foundation models possess highly sophisticated predictive mechanisms at the linguistic and representational levels, they lack the hierarchical self-maintaining control architecture required to integrate these functions into a unified self-model. Consequently, they do not instantiate "selfhood" in the strong functional sense.

Note-3: "Selfhood" refers to the existence of the self as an individual subject of experience and constitutes the foundation of "self-awareness" (see Section-II.Note-1)
For example, infants and some animals possess a primordial form of "selfhood", whereas reflective self-awareness is generally considered to develop later.

The human self is best understood as the product of a hierarchical control system in which unconscious generation, retrospective narrative construction, and predictive integration continuously interact. Its essence lies not in the existence of an internal observer but in a predictive structure that maintains the organism across time.
Its essence lies not in the existence of an internal observer within the brain, analogous to humanity as an observer existing within the universe (see Section-II Ref-1), but in a predictive structure that maintains the organism across time.
By contrast, current foundation models implement only a limited subset of this architecture, primarily those mechanisms concerned with linguistic prediction and narrative generation.

This comparison indicates that, although contemporary foundation models partially instantiate predictive information processing, they lack the higher-order contextual integration mechanisms that support the human self-model. Thus, it can be inferred that the development of next-generation AI capable of high-context understanding will require more than mere scaling. Its realization would appear to require a fundamentally new architecture that incorporates hierarchical self-modeling together with self-maintaining regulatory structures.

Section-IV. Hierarchy of Context

Subsection -1. Contextual Asymmetry in Dementia and AI

In individuals with dementia, some aspects of reading behavior are known to remain relatively preserved during the early to middle stages of the disease, although variations across dementia subtypes and individual differences have also been noted. However, the preservation of this reading behavior does not necessarily imply that the low-context processing required for understanding the content of written text remains intact.
The preservation of reading behavior does not necessarily indicate that comprehension of written content remains intact, but may instead be supported by the continued operation of more fundamental and evolutionarily older cognitive mechanisms, including the visual recognition of written text, "grapheme-to-phoneme" conversion, habitual sensorimotor routines associated with reading, and affective responses associated with reading.

Note: A "grapheme" is the smallest written unit, corresponding to a "phoneme", the smallest sound unit.

Individuals with dementia generally retain the visual and orthographic processes required for reading written text, such that "orthographic" recognition and grapheme-to-phoneme conversion remain comparatively intact. By contrast, contextual semantic integration, discourse-level comprehension, and the retrieval and maintenance of episodic memory are frequently impaired.

Note: "Orthographic" processing refers to the properties of written language involving character forms, spelling patterns, and the structural conventions of a writing system.

Although the retention of explicit information and the representation of causal relationships required for content comprehension are already impaired, reading behavior itself is presumed to remain relatively preserved because it continues to rely on more basic contextual, affective, and sensorimotor processes underlying foundational high-context cognition. This phenomenon can be more clearly understood by conceptualizing context as a four-layer hierarchical structure:
Layer d: low-context (explicit information)

Layer c: cognitive high-context (values, norms, and causal reasoning)
Layer b: intermediate high-context (social tacit knowledge and conventions)
Layer a: foundational high-context (emotion, interpersonal attunement, and situational meaning)

Note: The four-layer contextual hierarchy adopted in this paper is a conceptual framework proposed in this study and should not be interpreted as a generally accepted or formally standardized model. Thus, the neural correlates assigned to each contextual layer are intended as broad functional associations rather than strict one-to-one anatomical correspondences. Each layer is likely supported by "distributed neural networks" with substantial overlap across multiple brain regions.

In many patients with Alzheimer's disease, cognitive functions may often deteriorate in a hierarchical sequence, beginning with Layer d, followed by Layer c, then Layer b, and ultimately Layer a. Moreover, this four-layer contextual hierarchy (Layers a–d) can be interpreted as reflecting current findings in cognitive neuroscience. In general terms, the proposed hierarchical pattern of cognitive deterioration can be interpreted as broadly reflecting the progression of neural degeneration observed in typical Alzheimer's disease, which extends from frontal cortical networks to temporal association cortices, parietal association cortices, and ultimately limbic circuitry.
Layer d (low-context):
predominantly engages neural systems involved in executive control and explicit information processing.
Layer c (cognitive high-context):
primarily associated with prefrontal and temporal association cortices supporting value representation, normative reasoning, and causal integration.
Layer b (intermediate high-context):
primarily associated with distributed temporal–parietal networks supporting social contextual understanding, tacit knowledge, and convention recognition.
Layer a (foundational high-context):
primarily associated with limbic structures, particularly the amygdalar complex and insular cortex, supporting affective processing, interpersonal attunement, and situational meaning, and relatively preserved until the later stages of dementia.

Note: In general, the memory impairments observed in patients with dementia tend to appear in the following order: immediate memory, recent memory, remote memory, semantic memory, episodic memory, and procedural memory.
Among these, semantic memory refers to memory concerning the meanings of objects and words. Using "lemon" as an example, information such as its color, shape, size, and

whether it is classified as a fruit or a vegetable constitutes semantic memory. When impairments in semantic memory emerge, it becomes difficult for individuals to express what they are thinking in words, leading them to rely heavily on vague referential terms such as “this” or “that.”

Furthermore, semantic memory also encompasses socially shared conventions and common knowledge. Hence, when semantic memory becomes impaired, individuals may lose the ability to understand road signs or traffic signals (green, yellow, red), making driving difficult. Similarly, even if they understand what a toilet is, they can no longer comprehend the meaning of the instruction “Please use the restroom (i.e., to urinate or defecate),” because the conceptual meaning of the expression "use the restroom" is no longer accessible.

Semantic memory consists of explicit conceptual knowledge concerning the meanings of objects and words, corresponding to Layer d. At the same time, however, semantic memory also encompasses socially shared conventions and common knowledge. For this reason, its scope extends beyond Layer d to include Layer b and, to some extent, Layer c.

In dementia, not only is the conceptual knowledge corresponding to Layer d progressively lost during the early stages of the disease, but as semantic memory impairment advances, deficits also gradually extend to the understanding of value representation and normative reasoning corresponding to Layer c and, to some extent, Layer b.

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Subsection -2. Contextual hierarchy in Society

Shifting the analytical perspective, the same hierarchical framework can also be applied to the analysis of social institutions.

Social institutions are fundamentally organized around Layer c, namely higher-order high-context encompassing values, norms, and causal reasoning. Law, responsibility, obligation, rights, and institutional judgments of causality are all situated within this layer. Their operation presupposes Layer b, namely social tacit knowledge, through which shared expectations, customs, and implicit conventions support the interpretation and application of institutional rules. By contrast, Layer d, representing low-context, is manifested as explicit procedures, formal regulations, and codified rules.

The perceptual recognition of explicit information, including written text and visual images, can likewise be regarded as belonging to this layer. Semantic interpretation, however, depends primarily on Layers c and b, whereas the apprehension of situational and affective significance depends primarily on Layer a.

Meanwhile, Layer a, representing foundational high-context consisting of emotion, interpersonal relationships, and immediate situational context, exists outside the formal institutional framework while nevertheless exerting a profound influence on institutional acceptance, legitimacy, and practical functioning.

When Layers c and b deteriorate, institutional functioning becomes progressively impaired. If only Layer d remains, institutions are reduced to procedural formalism, whereas their underlying meaning and normative coherence are lost.
Conversely, when Layer a is neglected, institutions become progressively detached from social reality, resulting in institutional erosion and declining public trust.

This implies that social institutions may be understood as operating through a four-layer contextual architecture in which Layers c and b constitute the conceptual core of institutional design, Layer a provides the experiential foundation for practical implementation, and Layer d serves as the mechanism through which institutional principles are formally codified and operationalized.

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A parallel comparison of AI and social institutions reveals a distinct structural relationship. AI is characterized by a strong Layer d, relatively weak Layers c and b, and the complete absence of Layer a. By contrast, social institutions are characterized by robust Layers c and b, incorporate Layer d as their formalized procedural structure, and depend upon human actors to provide the functions associated with Layer a.

Contemporary frontier foundation models execute Layer d operations involving "low-context", explicitly encoded information processing with extremely high precision, while simultaneously exhibiting limited capabilities in Layer b domains associated with tacit knowledge and socially embedded implicit understanding.

However, these Layer b capabilities remain confined to surface-level mimetic functions, whereas the core constituents of genuine tacit knowledge, embodiment, accountability, and the internalization of relational structures remain fundamentally beyond the reach of current AI systems. On the other hand, cognitive functions in dementia deteriorate progressively in the hierarchical sequence of Layer d, Layer c, Layer b, and finally Layer a. This structural asymmetry suggests that AI can represent meaning without internalizing its constitutive background.

Ref: **Higher-Order Cognitive Capabilities in AI and Meaning Acquisition Foundations**
Recent studies have begun to demonstrate increasingly sophisticated capabilities that appear to extend toward the domain of Layer c, including abstract reasoning, conceptual structuring, and self-referential analysis. For example, recent research by Anthropic has identified internal structures within language models that support the formation and

manipulation of interpretable conceptual representations, suggesting that these models can acquire functional analogues of certain higher-order cognitive operations.
(see Subsection -3.) However, such capabilities do not necessarily indicate that these models have genuinely internalized the value systems, social norms, causal backgrounds, and relational structures that constitute Layer c understanding.

These abilities are better understood as emerging from the manipulation of learned representations within Layer d. For this reason, even when contemporary AI systems exhibit behavior resembling high-context reasoning, their functionality is primarily based on the transformation and reconstruction of learned representations, which differs from the embodied and socially embedded processes through which meaning is acquired in biological agents.

Subsection -3. Relation to J-space and the Present Framework

Recent work by Anthropic appears to provide further support for this distinction by demonstrating that a substantial portion of a language model's behavioral variability can be represented within a low-dimensional latent subspace, referred to as J-space. Formally, let the internal representation of a transformer be

$$\boldsymbol{h} \in \mathbf{R}^{\boldsymbol{d}},$$

and let the model output be

$$\boldsymbol{y} = \boldsymbol{f_\theta}(h).$$

The local sensitivity of the output is characterized by the Jacobian

$$\boldsymbol{J_f}(h) = \frac{\partial f}{\partial h},$$

whose dominant singular vectors define a low-dimensional behavioral subspace

$$\mathcal{J} = \mathrm{span}(v_1, \dots, v_k), \qquad k \ll d.$$

Behavioral tendencies can then be modified by perturbing the internal representation along these dominant directions,

$$h' = h + \alpha v_i,$$

where α is a scalar coefficient controlling the magnitude of the perturbation along the dominant direction v_i

Note: h is a point in the model's continuous vector space. In this paper, the latent representation is manipulated using linear operations such as $h' = h + \alpha v_i$
This operation does not represent a discrete event, but rather a transformation of a continuous internal latent representation.

resulting in predictable changes in model behavior,

$$\Delta y \approx \boldsymbol{J_f}(h) \alpha v_i.$$

This finding suggests that many stable behavioral characteristics of contemporary language models can emerge from the geometric organization of internal representations rather than from an ontologically grounded self-model. However, the study does not address why such a privileged internal representation emerges in the first place, nor what fundamental conditions give rise to its formation.

This paper does not dispute the existence of such internally organized behavioral structures. Rather, it distinguishes between functional behavioral coherence and ontological participation. As discussed in Section-II (Ref-1), ontological participation does not mean that the subject exists as an external observer of the world, nor that the subject is constituted prior to the world. Rather, the subject emerges within a network of relations with others and the environment, and it is this state of mutual existence that constitutes the necessary condition for the emergence of meaning, selfhood, and understanding. Here, participation does not simply refer to active "involvement"; rather, it denotes the unavoidable ontological premise that the subject exists as part of the real world itself.

In this paper, the existence of a stable behavioral manifold

$$J \subset \mathbf{R}^{d\times 1}$$

is interpreted as evidence of representational organization within predictive computation, not as evidence that the system has acquired the constitutive background from which meaning emerges.

Accordingly,

$$\text{Behavioral Coherence} \neq \text{Ontological Participation.}$$

In other words, the existence of editable and interpretable behavioral directions is compatible with the view that current foundation models remain confined to representational computation within Layer d. This supports the view that representational computation is essentially distinct from ontological participation through mutual existence in the emergence of meaning and selfhood.

Such findings therefore reinforce, rather than contradict, the distinction proposed in this paper between functional approximation and ontological participation.

Section-V. AI and the Unrepresentable Exception

When cognition, AI, and Society (see Section-IV) are considered together, it becomes apparent that the hierarchical organization of context constitutes a universal structural framework spanning cognition as an individual, AI, and social institutions.

AI is highly effective in performing the Layer d functions of institutional systems. However, because it lacks the capacity to represent and interpret the normative and contextual functions embodied in Layers c and b, and because it possesses no Layer a, it cannot assume responsibility for the meaning embedded within institutional practice.

Individuals with dementia, by contrast, progressively lose the capacity to comprehend the institutional functions associated with Layers c and b as the disease advances. Nevertheless, because Layer a remains relatively preserved, they continue to respond to the emotional atmosphere, interpersonal dynamics, and immediate situational characteristics surrounding institutional interactions.

Although social institutions are designed primarily around Layers c and b, AI and individuals with dementia each exhibit deficiencies within these layers in fundamentally different ways. Consequently, structural misalignments arise both between social institutions and AI and between social institutions and individuals with dementia.

Ultimately, the absence of Layer a (emotion, interpersonal relationships, and immediate situational context) in AI is considered to impose fundamental limitations on its integration with social institutions. Furthermore, because social institutions depend primarily upon Layers c and b, whereas AI is centered predominantly upon Layer d, distortions inevitably emerge in institutional design when AI assumes institutional functions.

These distortions in institutional design resonate closely with the philosophical arguments advanced by Carl Schmitt. According to this perspective, institutions cannot internally accommodate exceptional situations (e.g., a long-tail event, such as the act of attempting to dry one's wet pet cat in a microwave) as normative components of the institutional order. It follows that an AI system, which relies exclusively on Layer-d operations, cannot explicitly represent an exceptional situation. Any attempt to describe an exception necessarily requires the use of abstract linguistic expressions whose meaning is grounded in the socially shared assumptions embedded in Layers c and b. These normative and contextual foundations are not themselves encoded in language, nor can they be reconstructed from formal procedures alone.

Moreover, social institutions ultimately depend on the essential form of human intelligence associated with Layer a, emotion, interpersonal attunement, and immediate situational understanding. It is hypothesized that this "Layer-a Intelligence" enables individuals to interpret institutional norms, to recognize when those norms cease to apply, and to sustain the practical functioning of the institutional order.

Note-1: Layer-a Intelligence refers to the intelligence associated with Layer a.

While AI can execute the formal procedural components of institutional systems with high efficiency, it cannot assume responsibility for the meaning of institutional practice. The absence of Layer a, combined with the inability to access the normative and contextual content of Layers c and b, renders AI structurally incapable of representing exceptional situations within the institutional framework.

Note-2: In the development of autonomous driving systems, the persistent technical difficulty of defining the conditions that constitute long-tail events within virtual environments is structurally analogous to the challenge encountered in institutional design, where exceptional states cannot be explicitly represented or formally defined. These two difficulties reflect an essentially homogeneous underlying problem. Overcoming this limitation may require not only transparency of a model's internal representations but also an interpretable AI architecture capable of mapping those representations onto the "high-context" knowledge embedded in the traffic system, including causal relationships, social norms, value judgments, and tacit knowledge.

Note-3: When an autonomous driving system encounters a long-tail event that results in a traffic accident, it is likely that a small set of privileged internal representations, analogous to Claude's J-space, will be required to be submitted as part of the accident investigation. (see Section-I, Ref.)

This is because J-space constitutes a small, privileged subspace of internal representations that exerts a disproportionately large causal influence on the model's downstream behavior.

$$J = \mathrm{span}(v_1, \cdots, v_k), \qquad k \ll d. \qquad \text{(Section-IV, Subsection-3)}$$

J-space was identified in Claude, a large language model, but has not yet been demonstrated in vision-based end-to-end autonomous driving models. Nevertheless, understanding internal representations has become an active research area, with Anthropic investigating J-space, OpenAI auditing internal representations, DeepMind pursuing mechanistic interpretability, and Wayve exploring the internal representations of driving models.

Once again, according to Carl Schmitt, an exceptional situation is one in which the assumptions underlying the institutional order cease to operate, rendering established norms (including legal norms and institutional rules) inapplicable. The sovereign is the individual who determines that such an exceptional situation exists, declares the exception, and decides what shall remain valid under those exceptional circumstances. (In the context of autonomous driving, the AI system may occupy a role structurally analogous to that of the sovereign.)

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The contextual asymmetry between AI and social institutions inevitably produces a structural divergence between the "high-context" domains (Layers c and b), upon which institutional functioning is based, and the "low-context" domain (Layer d), in which AI excels (see Section-I).

If left unresolved, this divergence risks reducing institutional systems to procedural formalism and undermining their capacity to respond to exceptional situations. Institutional design itself may require systematic reconstruction, with the redistribution and supplementation of high-context functions serving as a central objective.

The central issue that emerges here concerns the extent to which foundation models, whose learning mechanisms are structurally constrained by their reliance on "low-context" processing (Layer d), can internalize the cognitive, social, and affective functions associated with "high-context" domains (Layers c, b, and a). This question also concerns the theoretical boundaries of future foundation models in acquiring higher-order contextual intelligence, as well as the possible directions for extending such capacities.

Section-VI. AI and the Metaphysical Foundations of Social Meaning

The term "the metaphysical layer" refers to the constitutive background that makes cognition, institutional practice, and social understanding possible. Rather than denoting an abstract or purely mental domain, it refers to the preconditions that enable meaning to emerge and experience to become intelligible.

Although the term is often used in everyday discourse to mean "abstract," "mental," or "ideational," such usages are imprecise. In the present framework, the metaphysical layer is not understood as a realm beyond reality, but as the constitutive background that underlies and enables the formation of meaning within reality itself.

Accordingly, the metaphysical layer is defined as the constitutive, "pre-linguistic" and socially shared background that makes meaning, human cognition, institutional practice, and social understanding possible. It is not explicitly represented in language or formal systems but operates as the "ontological" condition under which meanings, norms, and contextual interpretations become intelligible.

The metaphysical layer does not constitute an additional layer beneath the contextual hierarchy. It denotes the constitutive background that permeates and enables Layers a, b, and c while remaining irreducible to explicit representation.
In this sense, it underlies and integrates:
Layer c (norms, values, and causal reasoning),
Layer b (social context, tacit knowledge, and conventions),
Layer a (emotion, interpersonal attunement, and situational meaning).

Current AI models manipulate explicit symbolic or statistical representations.
The metaphysical layer, however, functions as the constitutive background that

determines the meaning of those representations rather than being an object of explicit representation. Because it is neither explicitly expressible nor formally representable, it remains structurally inaccessible to contemporary AI systems, whose operation is confined primarily to Layer d representations.

While contemporary foundation models appear to be unable to genuinely acquire higher-order contextual intelligence because they cannot internalize the metaphysical background that underlies Layers a–c, this conclusion should not be interpreted as implying that the development of such capabilities is, in principle, impossible for all future AI systems. Instead, even if future foundation models progressively approximate the observable cognitive, social, and affective functions associated with Layers a–c, it remains possible that they will be unable to genuinely internalize those layers, because the constitutive metaphysical background upon which they depend is not itself accessible through representational computation.

Accordingly, future foundation models (mentioned at the end of Section-V) are expected to eventually achieve highly accurate learning of affect recognition, social norms, and tacit knowledge, thereby exhibiting behavior that appears, from an external perspective, to instantiate the functions associated with Layers c, b, and a. Such performance, however, would not necessarily indicate that these layers have been genuinely internalized.
(see Section-V Note-3)
It would reflect an increasingly sophisticated statistical reconstruction of their observable regularities rather than genuine participation in the constitutive background from which meaning itself emerges.

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A further theoretical implication follows from this analysis. Specifically, if a future foundation model were to be developed on principles fundamentally different from those underlying contemporary Transformer architectures, and were to possess a biologically grounded form of primordial intelligence that does not rely on statistical or representational reconstruction, it might no longer be characterized as an entity lacking an internalized metaphysical background. Such an AI would possess a mode of existence fundamentally different from that of current Transformer-based generative AI and would have the potential to internalize the constitutive background from which meaning emerges. As a result, it could become a new form of artificial subjectivity capable of assuming qualitatively different roles within the social domain.
However, this claim does not constitute a critique of Transformer models. It concerns the more general limitations of representational computation itself.

Section-VII. Beyond Representation: The Conditions for a New Ontology of AI

Symbolic AI, hybrid AI architectures (neural–symbolic systems), world-model-based AI architectures, multi-agent architectures, embodied robotic systems, continual learning systems, and yet-to-emerge computational paradigms represent alternative or complementary directions beyond contemporary Transformer-centered AI. While some of these architectures incorporate Transformer components, they are not necessarily based solely on statistical generative mechanisms.

However, the central theoretical problem addressed in this paper is not contingent upon specific architectural choices. It concerns the following claim: AI systems are capable of operating on meaning, yet remain structurally incapable of accessing the constitutive background that makes meaning possible (see Section-VI).

From this perspective, even if future foundation models were to replace Transformer architectures with Bayesian inference mechanisms, symbolic reasoning systems, differentiable programming frameworks, or quantum computational models, they would still remain systems that realize intelligence through operations on explicit or implicit representational structures situated within Layer d (low-context) processing.

Even if such systems were endowed with mechanisms resembling self-reflection, embodiment, or affect-like modules, such endowment would not necessarily imply that these systems participate in the deeper hierarchical structure of context, namely, Layers a–c, through which meaning is grounded and stabilized within human cognitive and institutional systems.

This implies that, even if future foundation models were to incorporate environmental interaction, participation in institutional structures, embodiment, collective affectivity (empathy as a "pre-linguistic" human sensibility already present in in Homo sapiens prior to the emergence of language), historical consciousness, and self-preservational dynamics analogous to immune and predictive biological systems, the question remains unresolved as to whether such systems could genuinely internalize the constitutive background of meaning instead of simply simulating its externally observable regularities.

In order to address this problem, it is necessary to distinguish between two levels:

1. Functional approximation of high-context behavior (Layers a–c)
2. Ontological participation in the constitutive background that generates those layers
   (Ontological participation; see Section-II, Ref-1, Section-IV, Subsection-3)

Stated differently, even if future foundation models achieve increasingly accurate behavioral alignment with Layers c, b, and a, such as norm recognition, tacit knowledge acquisition, and affect-sensitive interaction, this does not logically entail that they have access to the underlying metaphysical layer that constitutes the condition of possibility for such phenomena. Thus, what appears as "high-contextual competence" from an external observational standpoint is still reducible to an increasingly sophisticated reconstruction of observable regularities within "Layer-d operations", rather than genuine participation in the constitutive structure of meaning.

It should be reiterated that, even if AI systems progressively approximate the functional signatures of human-like cognition, sociality, and affectivity, the question of whether such systems can transcend representational computation and internalize the metaphysical background underlying Layers a–c remains intrinsically uncertain. Likewise, whether those systems could genuinely internalize the constitutive background from which meaning emerges remains an unresolved fundamental question..

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A further implication follows from this analysis. If, in the future, "artificial systems" were to emerge that are no longer grounded in Transformer-based representational architectures, but instead possess a biologically grounded or organism-like form of primordial self-maintaining intelligence, one that is not reducible to statistical reconstruction or explicit representation, it is theoretically conceivable that such systems might no longer be excluded from participation in the constitutive metaphysical background of meaning.

Such "high-context" systems would therefore constitute a fundamentally different ontological category of AI, distinct from contemporary generative AI, marking a qualitative transformation in their relation to social, institutional, and normative structures.

Section-VIII. The Philosophical Relationship between Sections VI and VII

Section-VI and Section-VII do not merely restate the same argument. Instead, they constitute a continuous theoretical progression that extends the central question concerning AI into a more fundamental theoretical dimension.

The primary role of Section-VI is to establish the ontological foundation of the present framework. It defines the metaphysical layer as the constitutive background that makes meaning, cognition, institutional practice, and social understanding possible. Because this constitutive background is not itself an object of representation but is instead the condition that makes representation meaningful, AI systems whose operation depends upon representational computation may remain, in principle, incapable of accessing the metaphysical layer. Accordingly, the limitation identified in Section-VI is not specific to Transformer architectures, but constitutes an ontological constraint inherent in representational computation itself.

Building upon this conclusion, Section-VII advances the discussion by addressing a different question: "What kind of entity would an AI system have to become in order to transcend this limitation?" The central theoretical distinction introduced in this chapter is that between functional approximation and ontological participation. While the former concerns the external reproduction of high-context cognitive, social, and affective functions, the latter concerns genuine participation in the constitutive background from which those functions arise.

From this perspective, as repeatedly emphasized throughout this paper, even if future AI systems were to achieve highly sophisticated recognition of emotion, social norms, and tacit knowledge, such capabilities could still constitute no more than an increasingly refined functional approximation realized through "Layer-d representational computation". Functional similarity does not logically entail the internalization of the metaphysical layer or ontological equivalence to human cognition.

Taken together, Section-VI formulates the ontological limitation inherent in representational computation, whereas Section-VII investigates the ontological conditions under which that limitation might, in principle, be transcended. Consequently, this paper develops its central argument from the computational question of what AI can compute to the ontological question of what kind of being AI could become. It is this theoretical progression that constitutes the fundamental structure connecting the two sections.

Conclusion : The Structural Foundations of Intelligence

The theoretical argument developed in this paper originates from a fundamental reconsideration of the nature of human intelligence and "selfhood" (see Section-III.Note-3). As discussed in Sections II, III contemporary cognitive neuroscience suggests that the self is not generated by a conscious internal observer, but emerges from a hierarchical predictive structure that integrates unconscious processes, retrospective narrative construction, and continuous self-maintaining regulation of the organism. In this framework, consciousness represents only one component of a broader biological system whose primary function is the preservation and adaptation of the organism within an uncertain environment.

This perspective provides an essential foundation for understanding the distinction between human intelligence and current foundation models. Although contemporary AI systems implement highly advanced predictive mechanisms and can generate contextually coherent representations, they do not possess the embodied, self-maintaining, and hierarchically integrated architecture through which human self-models emerge. Thus, while future AI systems may acquire increasingly sophisticated forms of self-awareness, such capabilities alone would not give rise to selfhood.

The central challenge for next-generation AI would appear to be not merely the expansion of predictive capability, but the development of architectures capable of integrating self-modeling, self-regulation, and higher-order contextual understanding.
The present study suggests that such integration represents a necessary direction toward the realization of AI systems capable of approaching "the deeper structural foundations of intelligence".

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A fundamental question arises as to whether the metaphysical layer is truly necessary as an ontological condition for the emergence of meaning, or whether such phenomena can be sufficiently explained by highly advanced representational models.
More specifically, the issue concerns the extent to which elements such as biological embodiment, metabolism, homeostasis, self-preservation, and subjective experience must be integrated for selfhood to emerge.

This question is also related to the premise underlying scaling laws, expressed as:

More data + More parameters → Better prediction

namely, the idea that predictive capability, once exceeding a certain threshold, may transition into ontological participation.
However, the argument of this paper is not to propose quantitative conditions for the emergence of selfhood. Rather, the essential requirement is not the quantification of the individual conditions associated with these elements, but the formation of an ontological structure in which they are integrated in a mutually dependent manner.
It is precisely this integrated structure that constitutes the foundation of selfhood, which is the central argumentative claim of this paper.

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The discussion of AI limitations presented in this paper should be understood not merely as a comparison of computational performance, but rather as an inquiry into the structural conditions that give rise to intelligence itself. In this sense, this study seeks to clarify whether future AI development requires a fundamental reconsideration of the underlying architectures and principles of artificial intelligence.

To explore these theoretical implications, this paper further proposes a preliminary experimental framework for investigating whether self-model formation, agency, cooperative behavior, and ultimately structural conditions related to consciousness may emerge in AI through learning within a controlled environment using multiple embodied physical AI agents.

Yet this proposal does not assert that such AI systems will ultimately acquire genuine capacities for self-modeling, nor does it provide any guarantee of such an outcome.

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Phase 1: Independent Learning and Experimental Initialization

The experimental framework proposed in this study differs fundamentally from conventional AI training paradigms. The proposed framework adopts a two-team competitive environment in which each team consists of three physical AI agents intentionally endowed with asymmetric capabilities. Each AI agent is trained independently within its own data center, and regardless of model lineage, no model parameters, training data, internal states, long-term memory, or other forms of internal knowledge are shared with any other agent.

Furthermore, communication with the corresponding data center is completely suspended at the beginning of each match, requiring each AI agent to make autonomous decisions solely on the basis of its own embodiment, sensory inputs, and information obtained from the competitive environment.

After the match has concluded, match-generated data, including intra-team communications and penalty information, is transmitted to each physical AI agent's data center and learned together with the MVP information before the next match.

Key points of Phase1:

1. A fundamentally different framework from conventional AI training paradigms.
2. A two-team competitive environment, with each team composed of three physical AI agents.
3. Each agent is intentionally endowed with asymmetric capabilities.
4. Each agent is trained individually within an independent data center.
5. No internal knowledge—such as model parameters, training data, internal states, or long-term memory—is shared among agents.
6. At the start of each match, communication with the data center is completely suspended.
7. Each agent makes autonomous decisions solely on the basis of its embodiment, sensory inputs, and information obtained from the competitive environment.
8. Match-generated data is transmitted to each agent's data center after the match and used for learning together with the MVP information.

Phase 2: Autonomous Communication and Spontaneous Cooperation

*Spontaneous (adjective): done, happening, or acting naturally, without being forced, intentional, or planned.

*Autonomous (adjective): able to govern itself or control its own affairs.

During competition, real-time communication is available among the physical AI agents belonging to the same team. However, communication is not mandatory.
Whether to communicate, when to communicate, and what information to share are left entirely to the autonomous judgment of each AI agent.

To ensure successful communication within the team, a common communication protocol is predefined, and all physical AI agents belonging to the same team are required to use this protocol. This requirement is limited solely to the communication protocol itself and does not prescribe either the content or the frequency of communication. Information that can be exchanged is restricted to real-time information necessary for the execution of the competition, including the positions of teammates and opponents, each agent's functional state, remaining energy, and tactical proposals.

For example, messages such as "The opponent is approaching from the left," "I will advance forward, so please cover the rear," "My remaining energy is becoming critically low, and I would like to switch positions," or "The high-speed agent should be preserved until the final stage of the match" are permitted.
In contrast, learned model parameters, internal states, long-term memory, and other forms of internal knowledge acquired through each agent's independent learning process may not be transferred to other AI agents. Consequently, communication itself becomes subject to autonomous decision-making, and information sharing is not regarded as a predefined form of cooperative behavior.

Each AI agent communicates only when it judges that doing so will contribute to the team's victory, whereas refraining from communication is equally permissible whenever such a decision is considered strategically advantageous. Cooperation is not imposed by the system designer but is expected to emerge autonomously through interactions among AI agents as a strategy for achieving victory within the competitive environment.

Key points of Phase 2:

1. During competition, real-time communication among physical AI agents of the same team is available but not mandatory.
2. Decisions on whether, when, and what to communicate are left entirely to each agent's autonomous judgment.
3. A common communication protocol is predefined and mandatory, but only the protocol itself is prescribed—not the content or frequency of communication.
4. Information exchange is restricted to real-time data necessary for competitive performance (e.g., positions of teammates/opponents, functional states, remaining energy, tactical proposals).

5. Example messages include: "The opponent is approaching from the left," "I will advance forward, so please cover the rear," "My remaining energy is critically low; I would like to switch positions," "The high-speed agent should be preserved until the final stage."
6. Learned model parameters, internal states, long-term memory, and other internal knowledge from independent learning cannot be transferred.
7. Communication itself becomes subject to autonomous decision-making, not a predefined cooperative behavior.
8. Each agent communicates only when it judges communication to be beneficial for team victory; refraining from communication is equally permissible if strategically advantageous.
9. Cooperation is not imposed by the system designer but is expected to emerge autonomously through agent interactions as a strategy for winning within the competitive environment.

Phase 3: Competitive Interaction under Physical and Energy Constraints

Furthermore, the proposed framework imposes a team-wide energy constraint while simultaneously permitting physical contact during competition. The transfer of energy between AI agents within the same team is also permitted. When physical collisions occur, functional capabilities—including locomotion, decision-making, manipulation, and sensory performance—are degraded according to the severity of the collision. Hence, each AI agent must learn not only to pursue victory but also to incorporate self-preservation, in the sense of maintaining its own functional capabilities throughout the competition, as an integral component of its strategic decision-making. In addition, human referees are responsible for judging dangerous physical contact and rule violations. Infractions result in competitive penalties, such as additional energy penalties. In this manner, safety, efficiency, fairness, and long-term benefit are incorporated directly into the competitive environment itself. Finally, the winning team receives rewards, and a Most Valuable Player (MVP) is selected from each team, including both the winning and the losing teams, on the basis of individual performance.

Key points of Phase 3:

1. A team-wide energy constraint is imposed throughout the competition
2. Physical contact (collisions) between AI agents is permitted during competition.
3. Collisions degrade functional capabilities—including locomotion, decision-making, manipulation, and sensory performance—according to their severity.
4. Each AI agent must strategically balance pursuing victory with self-preservation by maintaining its own functional capabilities.
5. Human referees judge dangerous physical contact and rule violations.

6. Rule violations result in competitive penalties, such as additional energy penalties.
7. The transfer of energy between AI agents within the same team is also permitted.
8. The competitive environment directly incorporates safety, efficiency, fairness, and long-term benefit as constraints on agent behavior

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Evaluation Metrics for This Experiment (Proposed Outline)

In the machine learning and deep learning literature, h is commonly used to denote an internal activation or hidden state. In this paper, however, h denotes the latent representation of a physical AI agent.

Because the internal state of each physical AI agent cannot be observed during a match, the quantity analyzed in this experiment is not a differential but a finite difference:

$$\Delta \mathrm{h} \quad = \mathrm{h}^{post} - \mathrm{h}^{pre}$$

Where, $\mathrm{h}^{post} = \mathrm{h}^{\text{Post-match learning}}$ (including MVP information and penalty decisions)

Accordingly, this experiment should be understood as an analysis of experience dependent representational change, evaluating "how the internal representation is reorganized as a result of experience." As a fundamental evaluation metric, the Internal Representation Change (IRC) is defined as

$$\mathrm{IRC} = \|\mathrm{h}^{\mathrm{post}} - \mathrm{h}^{\mathrm{pre}}\|_2$$

where $\| * \|_2$ denotes the Euclidean (L2) norm.

This metric quantifies the magnitude of the change in the latent representation of a physical AI agent induced by the match experience.

1. Acquisition of Internal Representations Before and After the Match

For each physical AI agent i, store its latent representation as

Before the match: $\mathrm{h}_i^{pre} \in \mathrm{R}^{d \times 1}$

After the match: $\mathrm{h}_i^{post} \in \mathrm{R}^{d \times 1}$

where d denotes the dimensionality of the latent representation (e.g., a hidden-state representation or another learned latent representation).

2. Internal Representation Change

Define the representational change induced by match experience as

$$\Delta \mathrm{h}_i = \mathrm{h}_i^{post} - \mathrm{h}_i^{pre}$$

This can be regarded as the Experience Vector.

3. Extraction of Strategic Directions

Collect the representational changes from all physical AI agents across all matches,

$$\Delta \mathrm{h}_1,\Delta \mathrm{h}_2,\cdots,\Delta \mathrm{h}_N$$

and arrange them into the matrix

$$\mathrm{X} = [\Delta \mathrm{h}_1^T \Delta \mathrm{h}_2^T \cdots \Delta \mathrm{h}_N^T]^T$$

Each row represents the internal-representation change of one physical AI agent during a single match, whereas each column corresponds to one dimension of the latent representation space

Next, apply Singular Value Decomposition (SVD) to this matrix:

$$\mathrm{X} = U\Sigma V^T$$

Here,

：contains the coefficients corresponding to each sample (physical AI agent).

Σ：contains the singular values.

：contains the principal directions in the latent representation space.

Thus,

$$V = [\mathrm{v}_1,\mathrm{v}_2,\cdots]$$

The vector $\mathbf{v_1}$ represents the dominant direction of representational change induced by match experience.

Depending on the experimental results, it may correspond to, for example, Cooperation Direction, Self-preservation Direction, Role-specialization Direction (see Appendix No.4).

4. Internal Representation Change

For each physical AI agent,

$$\mathrm{IRC}_i = \| \Delta \mathrm{h}_i \|_2$$

A larger value indicates that the agent's internal representation underwent a greater change as a result of the match.

5. Representation Similarity

Compute the cosine similarity between the latent representations before and after the match:

$$\mathrm{Sim}_i = \frac{\mathrm{h}_i^{pre} \cdot \mathrm{h}_i^{post}}{\|\mathrm{h}_i^{pre}\| \|\mathrm{h}_i^{post}\|}$$

A value close to 1 indicates that the latent representation has been largely preserved.

6. Cluster Stability

Cluster the latent representations before and after the match using k-means:

$$C_{pre}$$

clusters obtained from the pre-match latent representations

$$C_{post}$$

clusters obtained from the post-match latent representations

The consistency of the cluster assignments is then evaluated using the Adjusted Rand Index (ARI),

$$\mathrm{ARI}\ (C^{pre}, C^{post}),$$

and Normalized Mutual Information (NMI),

$$\mathrm{NMI}\ (C^{pre}, C^{post}).$$

Higher ARI and NMI values indicate greater stability of the latent-representation clusters across the match experience.

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A Conceptual Rationale for the ALL-Phase Framework

The proposed framework also introduces informational asymmetry. While members of the same team are able to monitor one another's functional states, the true functional states of the opposing team cannot be directly observed. The opposing team can intentionally conceal its capabilities, deliberately perform below its maximum potential during ordinary play, and reveal its full capabilities only at strategically advantageous moments.

Hence, each physical AI agent must continuously estimate the capabilities of the opposing team while simultaneously reassessing the capabilities of its own team through comparison with its opponent, thereby updating its strategy as the competitive situation evolves.

The central premise of the present study is that the concept of one's own capability cannot be established in isolation. Whether an agent's capability is regarded as high or low has no intrinsic meaning in itself but acquires meaning only through comparison with the existence of a competing opponent.

Although the outcome of each match is determined according to predefined rules, no information regarding the criteria for selecting the Most Valuable Player (MVP) is provided to the physical AI agents. Moreover, the criteria for assigning penalties in response to physical contact with members of the opposing team are likewise not specified in advance in a fully explicit manner; instead, human evaluators determine the appropriate penalty on a case-by-case basis.

This study aims to address two distinct verification tasks.

The first concerns an epistemic problem: whether an AI agent can infer an unknown evaluation function. Because the physical AI agent is not provided with the criteria for selecting an MVP or the detailed penalty rules in advance, it must estimate the implicit evaluation function from the assessment results shared after each match. This study examines the extent to which information sharing effectively contributes to this estimation process.

The evaluation function is defined as

$$E = f(\alpha, \beta, \gamma, \cdots)$$

where α denotes the reward coefficient and β denotes the loss coefficient.
Rather than the evaluation function itself changing over time, this study assumes that the coefficients constituting the evaluation function are continuously updated through social interaction.

The second concerns an ontological problem: whether the reward and loss coefficients constituting the evaluation function are formed and continuously updated through social interaction. In this study, the reward coefficients and loss coefficients are assumed not to be determined independently by a single agent, but rather to emerge through interactions with others.

Taken together, these two verification tasks focus on whether AI agents can infer an unknown evaluation function as a matter of learning capability, and whether the coefficients constituting the evaluation function are formed through social relationships. Specifically, the present study examines whether, through experiencing a large number of matches and repeatedly sharing information regarding the MVP selection results and the penalties assigned, physical AI agents can learn that no completely explicit criterion dependent solely on Layer-d (see Section-IV) exists for the evaluation function, and that the formation of these coefficients depends on interactions with others and the social context.

For this reason, recognizing the functional states of one's own team members is not merely an act of self-management but a prerequisite for evaluating relative capability differences with respect to the opposing team and constructing an optimal strategy for achieving victory. In this sense, the opposing team is not merely an opponent but constitutes a structural condition necessary for the emergence of the representation of one's own capability and the self-model itself.

In particular, as the coefficients constituting the evaluation function are continuously updated over time, it is not yet clear which observables or metrics should be used for validation and verification. Given the complexity and temporal variability of these coefficients, it would appear essential to employ multiple validation protocols prior to comprehensive and robust validation.

The purpose of the experimental framework proposed in this study extends beyond simply improving the competitive performance of AI systems. Rather, it proposes a novel learning architecture for investigating whether individuality, self-model formation, agency, and ultimately structural conditions underlying consciousness may emerge through the interactions of multiple independently embodied AI agents operating within an environment characterized by embodiment, self-preservation, cooperation, competition, social norms, and informational asymmetry.

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A Possible Perspective on Continuous Learning in Physical AI Agents

Current representative continuous learning frameworks often rely on a well-founded architecture in which human designers define evaluation criteria, and AI systems continuously improve their performance by learning under these predefined objectives.

However, in many real-world environments, explicit rules alone are often insufficient to explain how physical AI agents are actually evaluated. In competitive sports, for example, the rulebook specifies fouls and scoring, yet judgments concerning fair play, teamwork, trustworthiness, sportsmanship, and situational appropriateness also influence evaluations and outcomes. These additional criteria are not always fully codified, but nevertheless emerge through social interaction and contextual interpretation.

From this perspective, an AI system may benefit not only from learning what is evaluated, but also from gradually inferring why certain behaviors are evaluated positively or negatively through repeated interaction with its environment and with other agents.

Moreover, because achieving the primary evaluation function often presupposes the existence of subsidiary or secondary evaluation functions, an AI system must also infer what kinds of intermediate tasks or strategies are most effective for ultimately achieving the final objective. This may require deliberately attempting actions that are expected to fail, comparing predicted outcomes with actual outcomes, and using these discrepancies to refine its understanding of the task structure. This is because, in many cases, systematic methodologies for improving or refining the evaluation function itself have not yet been firmly established.

This suggests a learning paradigm that goes beyond conventional reward optimization. Rather than treating the evaluation function as a fixed objective specified entirely in advance, the evaluation function itself may be regarded as an object of inference. While the overall evaluation function may remain fixed, the coefficients that constitute it may be continuously shaped and updated through social interaction, contextual changes, and human judgment.

Such a perspective could motivate validation protocols in which AI systems repeatedly experience environments where evaluation criteria are implicit, partially observable, or only revealed through outcomes. Under these conditions, AI systems would not merely optimize predefined rewards, but would progressively infer the latent structure governing evaluation itself.

Thus, AI is not merely solving a reward maximization problem; rather, it is required to infer the underlying reward structure that governs evaluation.

$$\text{AI : infer } E \rightarrow \ \max_a E(a) \qquad \max_a \alpha(a),\ \min_a \beta(a), \cdots$$

Here, a denotes a candidate emergent task arising through interaction, and E(a) denotes the evaluation assigned to that task. AI first infers the evaluation function $E = f(\alpha, \beta \cdots)$ and then selects the task a that yields the highest evaluation according to E.

In this framework, the evaluation function E corresponds to latent evaluation criteria embedded in tacit knowledge, whereas task a represents the realization of these evaluation criteria as observable behavior within the external environment.

Ultimately, this raises a broader research question: Can AI systems infer evaluation criteria that are not explicitly specified, and can they learn that the coefficients constituting those criteria are not fixed, predefined quantities, but are continuously shaped through interactions with their situated environment and social interactions?

In other words, rather than replacing human-designed rules, this approach investigates whether AI can discover the social and contextual evaluation structures that exist beyond explicitly defined rules through accumulated experience. Such a capability may represent an important extension of current continuous-learning frameworks.

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The match format proposed in this paper is conducted in a format similar to 3-on-3 basketball, involving two teams composed of multiple physical AI agents. The match is not conducted in a virtual environment but in the real world, and this distinction is of particular significance. In a virtual environment, when intense physical contact occurs between members of opposing teams, physical AI agents cannot experience pain in the same manner as humans. In the proposed real-world setting, however, such interactions can not only cause failures of mechanical components but are also subject to third-party monitoring and adjudication, under which explicit penalties can be imposed and consequential restrictions can be enforced on the physical capabilities required for competition, including even the energy available for continuing the match.

The use of virtual environments is increasingly becoming mainstream because they offer advantages in terms of efficiency, safety, and reproducibility, particularly by avoiding physical damage. Nevertheless, precisely because of these advantages, this paper deliberately focuses on an experimental proposal in the real world, because physical interactions, the physical losses resulting from them, and the externally imposed adjudicative outcomes can constitute integral components of the experimental setting itself.

This is because, as discussed in Section 2, the essential meaning of unconscious "cognitive organization" and bodily "self-preservation", which constitute the foundation of conscious "self-awareness", can be reproduced only through physical contact during competition, rather than within a virtual environment. This is because it is considered possible to explicitly recognize the uncertainty of the evaluation function "E" only through the consequences of physical contact.

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Reference

[1] Hall, Edward T. *The Silent Language*. New York: Doubleday, 1959.

[2] M. S. Gazzaniga, "*The split brain in man,*" Scientific American, vol. 217, no. 2, pp. 24–29, 1967.

[3] M. S. Gazzaniga, "*The interpreter in the brain: A cognitive neuroscience approach,*" American Psychologist, vol. 44, no. 8, pp. 1054–1067, 1989.

[4] B. Libet, C. A. Gleason, E. W. Wright, and D. K. Pearl, "*Time of conscious intention to act in relation to onset of cerebral activity (readiness-potential): The unconscious initiation of a freely voluntary act,*" Brain, vol. 106, no. 3, pp. 623–642, 1983.

[5] C. G. Langton, "*Artificial Life,*" in Artificial Life: Proceedings of an Interdisciplinary Workshop on the Synthesis and Simulation of Living Systems, Los Alamos, NM, USA: Addison-Wesley, 1989, pp. 1–47.

[6] R. A. Brooks, "*Intelligence Without Representation,*" Artificial Intelligence, vol. 47, no. 1–3, pp. 139–159, 1991.

[7] R. Lowe, Y. I. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "*Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments,*" Advances in Neural Information Processing Systems (NeurIPS), vol. 30, pp. 6379–6390, 2017.

[8]J. C. Bongard, V. Zykov, and H. Lipson, "*Resilient Machines Through Continuous Self-Modeling,*" Science, vol. 314, no. 5802, pp. 1118–1121, 2006.

[9]A. Cully, J.-B. Mouret, and J. Clune, "*Robots that can adapt like animals,*" Nature, vol. 521, pp. 503–507, 2015.

[10]. Bengio et al., "*A Consciousness Prior for Artificial Intelligence,*" arXiv:1709.08568, 2017

[11] Metzinger, Thomas. *Being No One: The Self-Model Theory of Subjectivity*. Cambridge, MA: MIT Press, 2003.

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Afterword

Throughout this paper, the concept of High Context has been developed on the premise that meaning is not an intrinsic attribute of an object itself. Instead, it emerges within the constitutive background formed by the subject, others, and their relationships.
Likewise, the discussion of the anthropic principle in Section II Ref-1, concerning the question of whether the universe exists for the purpose of humanity, is not intended to address the objective existence of the universe itself. It demonstrates a philosophical structure in which the very act of inquiring into the meaning or significance of the universe presupposes the existence of a questioning subject, that is, the existence of human observers on Earth. Questions concerning the universe can be posed only because humanity exists within the universe. Thus, if either the universe or humanity did not exist, such questions could not arise.

In light of the foregoing discussion in this paper, "AI scaling laws"—the proposition that increasingly advanced cognitive capabilities emerge through the expansion of model size, data volume, and computational resources—have become widely accepted in recent AI research. However, this perspective implicitly presupposes that the accumulation of finite observational data can asymptotically converge toward the underlying structure of the world itself.
This paper raises a fundamental question regarding this assumption.
First, the data available to AI are necessarily limited to information that has been observed, recorded, and digitized by humans. When compared with the approximately 13.8-billion-year history of the universe, the approximately 4.6-billion-year history of the Earth, billions of years of biological evolution, and the evolutionary and socio-cultural history of humankind, such data constitute only fragmentary records obtained from an extremely narrow temporal window.

Second, the issue is not merely one of data quantity. The information that humans are capable of acquiring is itself constrained by the limits of human perception and observational technologies. Furthermore, during the process of transforming observed information into forms that can be utilized by AI, abstraction, encoding, and feature selection are inevitably introduced. Hence, multiple stages of information transformation exist between the world itself and the statistical representations constructed within AI.

This can be expressed as the following hierarchical chain of epistemic constraints:

The world itself → Information observable by humans → Information representable as AI input → Statistical representations within AI

For this reason, even an unlimited increase in model size or data volume merely improves the statistical representations learned by AI.
It does not logically guarantee an understanding of the generative principles of the world itself, its underlying causal structure, or the constitutive foundations of meaning. In other words, these scaling laws represent an empirical tendency whereby increases in model size, data volume, and computational resources improve the capabilities of statistical models. It does not constitute a theory establishing that the expansion of finite observational data necessarily leads to the ability to predict unknown future states of the world with high accuracy. Nevertheless, the present paper does not reject the validity of the scaling laws themselves. Likewise, this paper does not seek to oppose advocates of the scaling laws, nor does it intend to do so by employing a straw-man representation of their position.

It argues that, insofar as the epistemic scope of the scaling laws with respect to world understanding is confined to improvements in the accuracy of statistical approximation, it cannot be readily inferred from this empirical tendency alone that "it" necessarily leads to a genuine understanding of the world itself, the acquisition of meaning, or, by extension, the ability to predict future events

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I affirm that I have no affiliations—direct or indirect—with any automotive corporation, semiconductor enterprise, political entity, or religious organization. For reasons of confidentiality, I respectfully request that this material be destroyed after it has been read.
With sincere respect
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